

Module 1 - AI Terms from readings

Key terms -- write your own definition after reading:

Term	Your definition (in your own words)
Foundational model	A foundational model is a basis for a general, broad-based AI system that can perform a variety of tasks and can later be adapted to complete more finite or fine tuned tasks.
Fine-tuning	Fine-tuning is a process where a model (that has already been trained) can be adapted for a specific task or more specific needs. Instead of creating an entirely new model, fine-tuning can reduce the time and cost with recreating a new model for a specific purpose.
In-context learning	In-context learning is a method of how a foundational model 'learns' what output is being requested by using examples given in the prompt versus using the inherent model parameters.
Emergence	Emergence is how the AI system behaves and responds is variable versus being intentionally constructed to respond in a well-defined, prescriptive manner. Because of this, how tasks are 'learned' and performed can produce unintended consequences can occur and cannot be anticipated.
Homogenization	Homogenization occurs when different methodologies and algorithms are used in machine learning systems where the output becomes more uniform and generic. This homogenization can present carry over problems including bias and flaws throughout multiple foundational models.

Term	Your definition (in your own words)
Benchmark	The use of a known data set to evaluate what a model has learned
Frontier model	Most advanced AI system that can be used for complex deep reasoning
Reasoning benchmark (e.g., MMLU, GPQA)	A test to see how well an AI system can accurately perform a complex task or solve a problem
AI adoption rate	How quickly AI is used in a set population
Compute scaling	The ability to adjust processing power according to the level of work required